



E&ICT Academy, IIT Kanpur

A Joint Initiative of MeitY & IIT Kanpur

Strategic Partner



CTO's Financial Toolkit (2) & Assessment Support

Session 8

Mentor:

Shubhashish Biswas

Introduction



Shubhashish Biswas

20 years experience in :

- **Managing Director**, Algo & Consulting Advisor, BCG
- **Practice Head, Digital & AI strategy** at KPMG, Accenture Strategy
- **VP (AI and Adv. Analytics)** at Citi
- **Product Management for Microsoft** (Machine Learning Offering on Microsoft Azure (Cloud))



POST GRADUATION

- **IIM Ahmedabad**
MBA (2-Year Full-time)
Focus: Technology / Systems & Finance
- **MIT, Cambridge (US),**
Pro. Masters Program
(Focus: Analytics & Data Science)

GRADUATION

IIT Madras
Bachelors Degree, Dept. of CSE
(Focus: AI & Analytics)



IIM
Ahmedabad
(2 Yr. PGP)



Massachusetts
Institute of
Technology
Cambridge, US



IIT Madras
Computer Science Dept.*



Session 8 Agenda



8.1

CTO's financial toolkit – A Refresher - continued

8.2

Tool Deep Dive & Mini Cases - continued

8.3

Capstone Project – AMA & guidance

The CTO's financial toolkit ecosystem

CXO

Enterprise Transformation

Driving organizational change through technology

AI Integration

Scaling AI adoption for maximum value

Risk Management

Mitigating cybersecurity threats and financial exposures

Business Strategy

Aligning technology with overall business goals

Cloud & Data Infrastructure

Optimizing cloud and data for efficiency and sustainability

Leadership Alignment

Unifying engineering, security, and finance teams

The Evolving CTO Role in 2026

Ref: [CTO Mandate in 2026 – CXO India](#)

Eight questions for the chief transformation officer



1

Do I have the full backing and confidence of the CEO and the board?



5

Am I coaching the CEO and top team and successfully changing how they lead the transformation?



2

Have I taken on incumbent vested interests and slaughtered relevant sacred cows?



6

Do I have a clear-eyed view on where the real value lies and where we cannot afford to compromise?



3

Have I established a rhythm and clock speed that is changing the metabolic rate of the business?



7

Have I intentionally picked some fights with senior line leaders and gotten them to change?



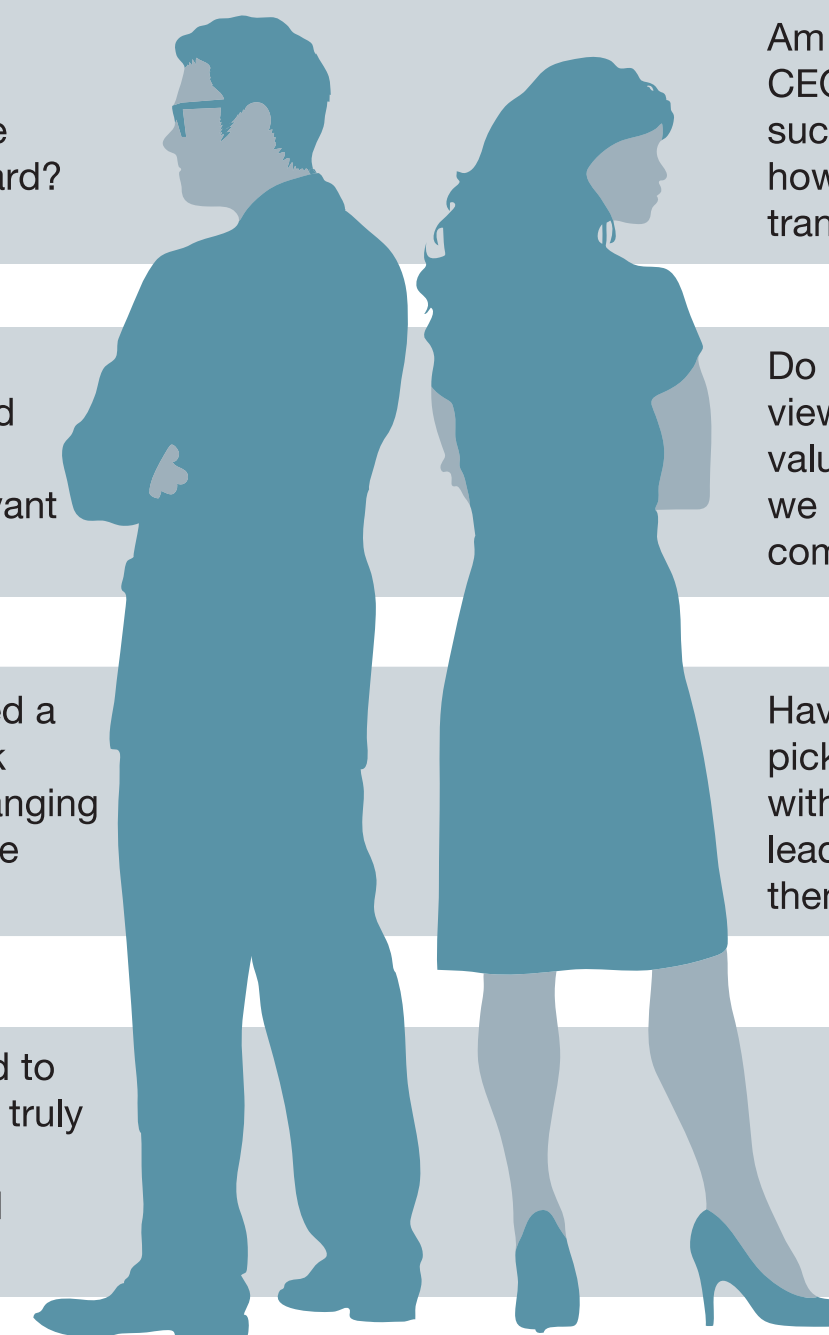
4

Have I connected to the front line and truly understood its perspectives and frustrations?



8

Do I understand the dominant culture and mind-set and where it needs to shift?



The CTO's financial toolkit ecosystem



Planning & Governance

- Budget Forecasting
- Investment Portfolio
- Chargeback Model



Investment & Spend Decision-making

- TCO Calculation
- Business Case Dev.
- Vendor Scorecard



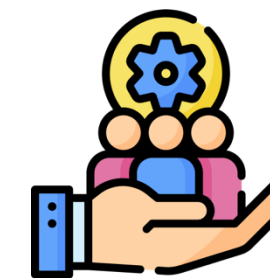
Operational Efficiency

- Cloud Spend (FinOps)
- Tech Debt Register
- Unit Economics



Risk Management

- Risk Register
- M&A Due Diligence



People

- Workforce Model

01

Budget Forecast



Background

PERSPECTIVE

In 2025–26, technology budgets face three simultaneous pressures: unplanned AI investment demands arriving mid-year, cloud spend swinging $\pm 30\%$ quarter-to-quarter, and CFO scrutiny that requires real-time justification of every dollar.

CTOs operating on a single annual budget submission lose credibility fast — they can't respond to board questions in real time, and they can't reallocate to high-priority initiatives without a protracted approval cycle.

A rolling 12–18 month forecast, refreshed monthly, changes the CTO's posture from reactive to strategic. It forces the team to own their numbers — increasingly a prerequisite for a seat at the executive table.

CORE ELEMENTS

1. Personnel & contractors (by team, role, FTE vs. contractor split)
2. Cloud & infrastructure (committed vs. on-demand breakdown, by product/environment)
3. Software licenses & SaaS (annualized, renewal dates flagged)
4. Security & compliance spend (explicit named category, not buried in IT overhead)
5. AI/ML platform costs (separate board-visible line item)
6. CapEx vs. OpEx classification (critical for EBITDA and accounting treatment)
7. Contingency reserve (10–15% of discretionary spend, pre-approved with a release process)



Tool Highlights

Budget Category	Q1	Q2	Q3	Q4	Annual Total	vs. Prior Year
PERSONNEL						
Engineering FTEs	\$1,200K	\$1,200K	\$1,280K	\$1,280K	\$4,960K	8%
Contractors & Staff Aug.	\$180K	\$200K	\$200K	\$180K	\$760K	15%
Training & Certification	\$40K	\$20K	\$40K	\$20K	\$120K	New
INFRASTRUCTURE & CLOUD						
Cloud Committed (RIs/Savings Plans)	\$420K	\$420K	\$420K	\$420K	\$1,680K	5%
Cloud On-Demand / Variable	\$120K	\$140K	\$130K	\$110K	\$500K	22%
Data Center / Co-location	\$80K	\$80K	\$80K	\$80K	\$320K	0%
SOFTWARE & SAAS LICENSES						
Productivity & Collaboration	\$60K	\$60K	\$60K	\$60K	\$240K	2%
Developer Tools	\$45K	\$45K	\$45K	\$45K	\$180K	10%
Security & Monitoring Tools	\$90K	\$90K	\$90K	\$90K	\$360K	20%
AI / ML PLATFORMS						
Foundation Model APIs	\$80K	\$100K	\$120K	\$130K	\$430K	New
MLOps & Data Platform	\$60K	\$60K	\$60K	\$60K	\$240K	New
SECURITY & COMPLIANCE						
Penetration Testing	\$50K	\$0	\$50K	\$0	\$100K	25%
Compliance Tooling (SOC2 etc.)	\$30K	\$30K	\$30K	\$30K	\$120K	10%
CONTINGENCY (12%)	\$280K	\$290K	\$305K	\$300K	\$1,175K	
TOTAL OPEX	\$2,735K	\$2,735K	\$2,910K	\$2,905K	\$11,285K	11%
CAPEX (Hardware, Licenses >1yr)	\$400K	\$0	\$200K	\$0	\$600K	-20%
TOTAL BUDGET	\$3,135K	\$2,735K	\$3,110K	\$2,905K	\$11,885K	9%



Mini Case



TechCorp is a 3,000-person B2B SaaS company. The CTO is presenting FY2026 budget to the CFO and CEO. Midway through FY2025, the board asked for an AI strategy — which now needs funding. CFO's opening position is: 'I want 5% cost reduction in tech overall.'

What is your strategy to set the budget forecast for it?

02

Investment Portfolio Tracker



Tool 2: Investment Portfolio Tracker

Introduction

PERSPECTIVE

The Run/Grow/Transform framework divides technology investment into three buckets:

- Keeping the lights on (Run),
- expanding existing capabilities (Grow),
- creating the future (Transform).

Most enterprises spend 70–80% on Run without realizing it, starving Grow and Transform.

CTOs who manage a technology investment portfolio rather than a project list make fundamentally better allocation decisions. They can defend 'why so much on infrastructure?' and 'why only \$200K on AI?' with data, not intuition.

In 2025, AI is the most contested item in Transform budgets. The portfolio lens helps CTOs allocate AI investment deliberately — not reactively — and show the board that technology strategy is coherent, not ad hoc.

CORE ELEMENTS

1. Run portfolio (infrastructure, licenses, support — keeping the business operational)
2. Grow portfolio (product enhancements, market expansion, process automation)
3. Transform portfolio (AI, platform re-architecture, new business models)
4. Current allocation vs. industry benchmark vs. target allocation (side-by-side)
5. Portfolio health metrics (ROI by initiative, run rate vs. budget, on-time completion)
6. Strategic alignment score (each initiative linked to a named board priority)
7. 18–24 month rebalancing plan (the path from current to target allocation)



Tool 2: Investment Portfolio Tracker

Tool Snapshot

Category	Current (\$K)	Current (%)	Benchmark	Target (%)	Target (\$K)	Action Required
RUN (Keep the lights on)						
Infrastructure & Cloud	\$2,800K	24%			\$2,400K	Reduce via FinOps (Tool 04)
Legacy System Maintenance	\$3,200K	27%			\$2,200K	Reduce via Tech Debt (Tool 05)
Support & Operations	\$1,400K	12%			\$1,200K	Automate to reduce
RUN SUBTOTAL	\$7,400K	62%	50–60% target	55%	\$6,490K	Free \$910K over 18 months
GROW (Expand existing capabilities)						
Product Feature Development	\$2,600K	22%			\$3,000K	Increase with Run savings
Process Automation	\$600K	5%			\$800K	Increase
Data & Analytics	\$400K	3%			\$600K	Increase
GROW SUBTOTAL	\$3,600K	30%	25–35% target	30%	\$3,540K	Hold / slight increase
TRANSFORM (Create the future)						
AI / ML Platform & Features	\$430K	4%			\$900K	Major increase — board priority
Architecture Modernization	\$320K	3%			\$600K	Increase as debt clears
New Business Model Experiments	\$50K	1%			\$270K	Create dedicated fund
TRANSFORM SUBTOTAL	\$800K	8%	10–20% target	15%	\$1,770K	Invest \$970K more/yr
TOTAL PORTFOLIO	\$11,800K	100%		100%	\$11,800K	Rebalance, not increase



Mini Case



ValNext Board Asks: 'Why Does Tech Cost So Much and Produce So Little?'

The board of a \$150M ARR SaaS company ValNext challenges the CTO: engineering headcount grew 30% last year but feature velocity feels flat. 'Where is all the money going?' The CTO has the board meeting in 10 days. What will you do in the 10 days?

03

IT Chargeback & Showback Model



Tool 3: Chargeback & Showback Models

PERSPECTIVE

When technology costs are pooled and invisible to business units, each team rationally maximizes its own consumption: spin up more resources, request more features, onboard more users — because the cost falls on IT's budget, not theirs. Chargeback makes each team accountable for what they consume.

An IT chargeback model is a financial system where an IT department tracks and bills internal business units for the specific technology resources and services they use. Instead of treating IT as a general company expense, this model shifts actual costs to the specific teams that create the demand.

An IT Showback model (reporting without billing) is a softer first step: teams see their monthly consumption costs, which alone drives 15–25% waste reduction even without a billing mechanism. For most organizations, showback is the right first step before full chargeback.

In 2025, AI cost governance is the newest and most urgent application: when an engineering team can run 10,000 foundation model API calls in an afternoon with no visible cost, they will. AI costs must be allocated by team and product — via showback at minimum — before they become uncontrollable.

CORE ELEMENTS

1. Service catalog (list of all IT services with a defined unit price for each)
2. Consumption tracking by business unit (cloud resources, licenses, support tickets)
3. AI usage tracking per team and per application (tokens, GPU hours, API requests)
4. Monthly allocation report (showback: 'your team consumed \$X this month')
5. Chargeback mechanism (how and when costs are transferred to business unit budgets)
6. Dispute resolution process (how business units challenge incorrect allocations)
7. Governance model (who sets service prices, how often reviewed, who adjudicates disputes)



Tool 3: IT Chargeback & Showback Models

Tool Snapshot

Service	Unit	Unit Price	Primary Consumer	Jan Usage	Jan Allocation	Notes
INFRASTRUCTURE						
AWS Compute (EC2 equivalent)	vCPU-hour	\$0.06	Product, Data	42,000 hrs	\$2,310	Tagged by team in cloud console
Cloud Storage (S3 equivalent)	GB-month	\$0.03	Data, Analytics	180,000 GB	\$4,500	Uncompressed storage
Managed Database (RDS equivalent)	DB-instance-hour	\$0.22	Backend Eng.	4,400 hrs	\$968	Per active instance
DEVELOPER PLATFORM						
GitHub Enterprise (per seat)	seat-month	\$21	Engineering	180 seats	\$3,780	Annual license amortized monthly
CI/CD Pipeline Minutes	compute-minute	\$0.01	Engineering	180,000 min	\$1,440	Build job minutes
Monitoring (Datadog)	host-month	\$28	All teams	85 hosts	\$2,380	Per active monitored host
AI / ML SERVICES						
Foundation Model API (GPT-4 class)	1,000 tokens	\$0.03	Product, CX	4.2M tokens	\$105	Tagged by team + feature
GPU Compute (training jobs)	GPU-hour	\$2.80	ML Team	320 GPU-hrs	\$896	On-demand GPU pool
SUPPORT SERVICES						
Help Desk (L1/L2 support)	ticket	\$45	All BUs	280 tickets	\$12,600	Attributed to requestor's BU
Security Reviews (per feature)	review	\$800	Product, Legal	4 reviews	\$3,200	Per feature security review
SAMPLE MONTHLY SHOWBACK: Product Team						
Infrastructure (compute, DB, storage)					\$4,820	
Developer platform (seats, CI, monitoring)					\$2,100	
AI / ML services (API + GPU)					\$3,640	
Support services (tickets raised)					\$1,350	
PRODUCT TEAM TOTAL — JANUARY					\$11,910/month	



Tool 3: IT Chargeback & Showback Models

Mini Case



In a mid-market tech products firm, three product teams are independently using foundation model APIs with no central visibility. The monthly AI API bill jumped from \$28K to \$112K in 90 days. No team knows how much they are spending. The CTO needs governance without bureaucracy that slows product development.

How will you govern this as the CTO?

04

Total Cost of Ownership (TCO) Calculator



Tool 4: TCO Calculator

PERSPECTIVE

CTOs routinely face decisions that look cheap in Year 1 but expensive by Year 3: an open-source tool requiring 2 dedicated engineers to maintain, a SaaS platform whose per-seat pricing explodes with growth, or an on-prem investment that can't scale without another data center build.

In 2025, the build vs. buy calculus has shifted again: AI tools reduce engineering effort, but also create new maintenance surfaces. The TCO model must now account for AI-augmented development costs, not just traditional FTE ratios.

A rigorous 5-year TCO protects the CTO from the most common trap: optimizing for the lowest Year 1 cost while ignoring the compounding cost of lock-in, integration debt, and scaling penalties.

CORE ELEMENTS

1. Acquisition / implementation costs (Year 0, one-time)
2. Annual operating costs (licensing, hosting, vendor support)
3. Internal engineering cost (integration + ongoing maintenance FTEs)
4. Scaling cost model at 2x and 5x current volume
5. Migration / exit cost (what it costs to leave the vendor or deprecate the build)
6. Opportunity cost of engineering hours diverted from product work
7. Risk-adjusted adjustments (vendor stability, technology obsolescence probability)

Tool 4: TCO Calculator

Tool Snapshot

TCO Calculator: Build vs. Buy — 5-Year View

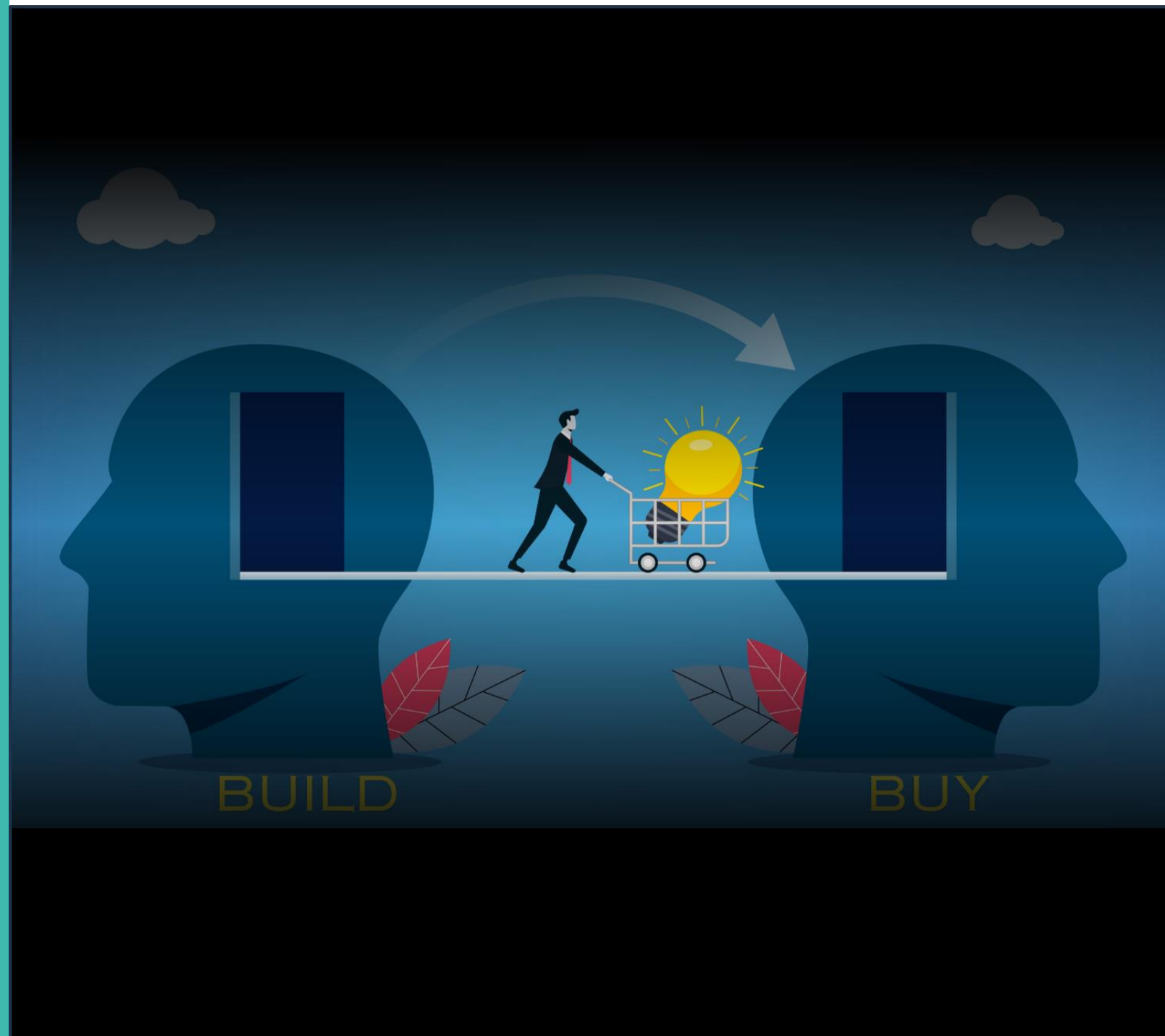
USAGE NOTES

- Opportunity cost = engineering hours × \$80/hr blended rate diverted from product features that generate revenue.
- Buy wins here due to high maintenance burden of custom data platforms. The calculus reverses for core IP that differentiates your product.
- Always include exit/migration cost in the Buy column — typically 6–12 months of engineering time to switch vendors.
- Recalculate annually: vendor pricing models change significantly, especially for AI-adjacent and data tooling.

Cost Component	Build (Internal)	Buy (Databricks+Snowflake)	Verdict
YEAR 0: IMPLEMENTATION			
Engineering time (build)	\$480K	\$80K	Buy wins
Infrastructure setup	\$120K	\$20K	Buy wins
Training & certification	\$40K	\$60K	Build wins
Total Year 0	\$640K	\$160K	Buy saves \$480K
YEARS 1–3: ANNUAL OPERATING (avg/yr)			
Licensing / SaaS subscription	\$0	\$480K/yr	Build wins (on paper)
Cloud compute costs	\$360K/yr	Included in license	Comparable
Engineering maintenance (FTEs)	\$640K/yr (2 FTEs)	\$80K/yr (0.25 FTE)	Buy wins decisively
Vendor support & SLA	\$0 (internal)	\$60K/yr	Build wins
Annual Operating Total	\$1,000K/yr	\$740K/yr	Buy saves \$260K/yr
YEARS 4–5: SCALE (5x volume)			
Additional engineering headcount	+\$640K (1 FTE)	\$0 (scales in license)	Buy wins
Infrastructure / compute scaling	\$400K	\$240K	Buy wins
License scaling (usage-based)	0	+\$180K	Build wins
5-YEAR TOTAL COST OF OWNERSHIP			
Total Cost (Year 0–5)	\$6,640K	\$4,560K	Buy saves \$2.08M
Engineering hours consumed	~18,000 hours	~2,500 hours	Buy saves 15,500 hrs
Opportunity cost of eng. hours*	\$1,440K	\$200K	Buy saves \$1.24M
RISK-ADJUSTED 5-YEAR TCO	\$8,080K	\$4,760K	Buy wins by \$3.32M



Tool 4: TCO Calculator Mini Case



Evaluating a Custom Data Platform vs. vendor-delivered Databricks + Snowflake

A \$200M ARR company's data engineering lead wants to build a custom lakehouse. A vendor team is pitching Databricks + Snowflake.

You as the CTO have 6 weeks to make a call and present to the CEO and CFO.

05

Business Case Builder



Tool 5: Business Case Builder

PERSPECTIVE

Technology initiatives fail to get funded not because they lack merit, but because they are presented in engineering language: story points, uptime percentages, and capability descriptions that mean little to a CFO. A structured business case translates technical investment into financial outcomes.

In 2025, two dynamics raise the stakes: AI investments require a business case because costs are visible (API spend, platform licenses) but benefits are diffuse; and higher cost of capital means payback periods matter more. An 18-month payback beats a 36-month payback even at higher NPV.

CTOs who master business case building earn a different kind of credibility — they become investment allocators, not just technology managers. That is the path to a real seat in strategic conversations.

CORE ELEMENTS

1. Executive summary (the problem, solution, the ask, and the payback period)
2. Investment required: one-time + ongoing costs, CapEx vs. OpEx clearly separated
3. Quantified benefits: revenue impact, cost savings, risk reduction — all in dollars
4. Payback period (months to break-even — the top metric for most boards in 2025)
5. Net Present Value (NPV) at the company's cost of capital / hurdle rate
6. Internal Rate of Return (IRR) — the annualized return on the investment
7. Sensitivity / stress test at 50% of forecast benefits
8. Go/No-Go decision criteria with named milestones and explicit stop triggers

Tool 5: Business Case Builder

Tool Snapshot

One-Page Business Case Template

For technology investments above \$250K, should be presented before detailed project planning begins, to get early executive alignment.

Usage Notes

- Always include a 50% stress-test scenario — it preempts the CFO's first pushback and shows the investment is robust.
- Payback period is the most important metric for most boards in 2025, given elevated cost of capital.
- NPV uses a discount rate equal to your company's hurdle rate — typically 8–15% for enterprise. Confirm with Finance.
- For AI investments, apply conservative benefit estimates — early AI project credibility is at stake if results underperform.

EXECUTIVE SUMMARY	
The Problem	Customer onboarding takes 14 days (industry avg: 3 days). We lose ~12% of new customers during onboarding. At \$1.2M new bookings/month, this is \$144K/month in churn.
The Solution	Automated onboarding platform: API integrations, guided workflows, ML-based at-risk detection, and real-time progress dashboards for customers.
The Ask	\$850K total investment in Year 1. Expected payback: 7 months (base case).
INVESTMENT REQUIRED	
Engineering (12 months, 3 FTEs)	\$480K
Third-party tooling & APIs	\$120K
Infrastructure (cloud)	\$60K
Project management & QA	\$80K
Contingency (15%)	\$110K
Total Investment	\$850K Year 1; ongoing cost \$120K/yr
QUANTIFIED BENEFITS	
Onboarding churn reduction (60% of current 12% rate)	\$86K/month → \$1,030K/year
Support cost reduction (fewer onboarding tickets)	\$15K/month → \$180K/year
Engineering time freed (fewer manual escalations)	\$25K/month → \$300K/year
Total Annual Benefit (recurring)	\$1,510K/year
FINANCIAL METRICS	
Payback Period	7 months (base) / 13 months (50% benefit stress test)
3-Year NPV (10% hurdle rate)	\$2,870K
Internal Rate of Return (3-year)	174%
Stress Test: benefits at 50% of forecast	Still pays back in 13 months — investment is robust
DECISION CRITERIA	
Go Trigger	Board approval + engineering capacity confirmed for Q1 start
6-Month Success Metrics	Onboarding time under 5 days; onboarding churn below 6%
Stop Trigger	At Month 3, if onboarding time not reduced 30%, pause and reassess scope



Tool 5: Business Case Builder

Mini Case



Proposing a \$3M AI Implementation to a Skeptical CFO

A financial services CTO wants to implement AI-assisted underwriting across three product lines. The ask is \$3M over 18 months. The CFO has seen overpromised AI projects fail at two previous companies. The board has requested an AI strategy but hasn't committed budget.

How should the CTO make the case?

06

Vendor Scorecard



Tool 6: Vendor Scorecard

PERSPECTIVE

The average enterprise runs 200–300 SaaS applications. Procurement decisions were made by different teams at different times, often without coordination. The result: overlapping capabilities, unused licenses, and auto-renewed contracts nobody remembers signing.

In 2025, the pressure is acute: CFOs are running SaaS rationalization audits in response to macroeconomic scrutiny, and the CTO is expected to lead the technical component. AI tooling (Zylo, Torii, Productiv) makes it possible to measure actual utilization vs. licensed seats in days.

Beyond cost savings, vendor scorecard discipline forces strategic clarity: if the company runs 5 different monitoring tools, that is not just waste — it is a signal of underdeveloped architectural standards. The vendor portfolio reveals strategy gaps hiding in plain sight.

CORE ELEMENTS

1. Complete vendor inventory (all contracts, annual spend, renewal dates, named tool owners)
2. Utilization rate (licensed seats vs. active users in last 90 days, by tool)
3. Strategic fit score (essential / useful / redundant / legacy)
4. Switching cost estimate (migration effort + productivity loss if consolidated)
5. Negotiation leverage assessment (competitive alternatives, market pricing)
6. Total SaaS spend as % of ARR (8–15% is typical for B2B SaaS)
7. Priority action matrix: renew / renegotiate / consolidate / eliminate

Tool 6: Vendor Scorecard

Tool Snapshot

Vendor / Tool	Annual Spend	Seats Licensed	Active Users (90d)	Utilization	Strategic Fit	Recommended Action
Salesforce (CRM)	\$480K	250	238	95%	Essential	Renew — negotiate 5% reduction
Slack	\$180K	800	720	90%	Essential	Renew at current terms
GitHub Enterprise	\$140K	200	185	93%	Essential	Renew at current terms
Datadog (monitoring)	\$210K	—	—	High	Essential	Renegotiate with New Relic competitive bid
PagerDuty	\$95K	150	48	32%	Redundant	Eliminate — consolidate into Datadog
Zendesk	\$120K	80	62	78%	Useful	Downgrade plan — save \$30K/yr
Confluence	\$90K	800	280	35%	Redundant	Review — Notion used by 400 users actively
Notion	\$75K	400	390	98%	Useful	Renew — migrate Confluence users here
Snowflake	\$320K	—	—	Medium	Essential	Optimize query costs — est. \$60K savings
Zoom	\$95K	800	750	94%	Essential	Renew at current terms
15 tools under \$50K each	\$420K	—	—	Mixed	Mixed	Audit individually — 30%+ elimination likely
TOTALS / SAVINGS SUMMARY	\$2,225K total					
Immediate savings (eliminate + downgrade)	-\$125K/yr					
Negotiation savings (competitive bids)	-\$85K/yr					
Query/usage optimization savings	-\$60K/yr					
TOTAL IDENTIFIED SAVINGS	-\$270K/yr (12% of SaaS spend)					

SaaS Vendor Audit & Optimization Scorecard

- Run annually before budget season.
- Prioritize by annual spend.
- Focus optimization on the top 20 vendors — typically 80% of total SaaS spend.

USAGE NOTES

- Utilization below 50% is a strong signal to consolidate or eliminate — below 30% is almost always waste.
- Bring competitive quotes to every renewal above \$100K — you do not need to switch, just demonstrate you could.
- SaaS spend benchmark: 8–15% of ARR for B2B SaaS. Above 15% warrants an immediate audit.
- Check for shadow IT: use expense report data and SSO logs to find unapproved tools teams are paying for personally.



E&ICT Academy, IIT Kanpur

A Joint Initiative of MeitY & IIT Kanpur

Strategic Partner



Tool 6: Vendor Scorecard Mini Case



\$2M in SaaS Budget Cuts Required by End of Quarter

The CFO announces company-wide cost reduction: every function must cut 15% of discretionary spend. Technology's target is \$2M over 12 months.

You, the CTO have 30 days to present a plan that does not cripple engineering velocity.

How will you plan it?

07

Technical Debt Quantifier



Tool 7: Tech debt quantifier

PERSPECTIVE

Technical debt is invisible on the balance sheet but highly visible in engineering velocity, incident frequency, and customer churn. 'We need to refactor this' is not a budget argument — 'this costs us \$2.4M per year in lost productivity and incident recovery' is.

CTOs who quantify technical debt financially win two things: budget for modernization that would otherwise never be approved, and a shared language with the CFO about why engineering is slower than headcount growth would suggest it should be.

In 2025, technical debt has a new dimension: AI readiness. Monolithic architectures, poor data quality, and missing APIs make it difficult or impossible to deploy AI capabilities. The cost of AI-unreadiness is now a board-level concern the CTO must be prepared to quantify.

CORE ELEMENTS

1. Debt inventory (system, area, severity, estimated fix effort in engineer-weeks)
2. Velocity tax (engineering hours lost per sprint working around debt items)
3. Incident cost (MTTR, incident frequency, blast radius cost per production incident)
4. Opportunity cost (features not built because capacity is consumed by debt-related work)
5. Risk cost (probability x impact of a debt-related production failure)
6. AI readiness penalty (debt that blocks or significantly increases AI adoption cost)
7. Annual cost of carry (total cost of maintaining debt in its current state)
8. Remediation ROI (cost to fix vs. ongoing annual cost to carry)

Tool 7: Tech debt quantifier

Tool Snapshot

Technical Debt Financial Impact Register

Maintained by Engineering Leads. Reviewed with CTO quarterly. Summarized for CFO annually as part of the investment portfolio review.

Usage Notes

- Annual Cost of Carry = (velocity tax hrs/yr × blended eng. rate) + (incident MTTR × frequency × \$/hr) + opportunity cost of blocked features.
- The full remediation program pays back in ~5 months: \$1.24M investment eliminates \$3M/yr of productivity drag.
- Always add an AI-readiness assessment: 'This architecture requires an additional \$2M re-platform before AI features can be deployed here.'
- Use this register in quarterly board updates: debt reduction is a leading indicator of future velocity improvement.

Debt Item	System	Severity	Annual Cost of Carry	Fix Cost	Payback	Priority
Monolithic payment service (no service boundaries)	Core Platform	Critical	\$840K/yr	\$320K	4.6 months	P0
Legacy auth system (no SSO, manual provisioning)	Identity	High	\$380K/yr	\$180K	5.7 months	P1
Manual data pipeline (Python scripts, no orchestration)	Data Platform	High	\$520K/yr	\$240K	5.5 months	P1
Undocumented API layer (3rd-party integrations)	Integrations	Medium	\$290K/yr	\$140K	5.8 months	P2
Test coverage under 30% on billing module	Billing	High	\$430K/yr	\$160K	4.5 months	P1
Deprecated Node.js 14 across 8 services	Infra / Security	Critical	\$240K/yr	\$80K	4.0 months	P0
No observability on 60% of production services	Observability	Medium	\$320K/yr	\$120K	4.5 months	P2
PORTFOLIO TOTALS			\$3,020K/yr	\$1,240K	~5 months avg	
NET ROI OF FULL REMEDIATION			Eliminates \$3M/yr drag	\$1.24M investment	Payback: 5 months	IRR: 200%+



Tool 7: Tech debt quantifier

Mini Case



Making the Case for a 6-Month Platform Modernization

Engineering is 40% slower than two years ago due to accumulated technical debt. The CTO wants to dedicate 6 months and 8 engineers to a modernization sprint — roughly \$1.5M in diverted capacity.

The CEO responds: 'That is \$1.5M for something customers will never see. Convince me.'

08

Unit Economics based Cost Allocator



Tool 8: Unit Economics based cost allocator

PERSPECTIVE

Unit economics — the cost and revenue attributable to a single unit of value — are a standard expectation in board and investor conversations. When the CFO asks 'what is our technology cost per customer?' the CTO needs an immediate, data-backed answer.

Without unit economics, pricing decisions, customer tier structures, and infrastructure investment decisions are made on intuition. With them, the CTO can model margin impact of growth, identify customers served at a loss, and make the case for pricing adjustments.

In 2025, AI costs are reshaping unit economics dramatically: an AI-powered feature can cost \$0.04–\$0.15 per user interaction — orders of magnitude more than serving a static web page. CTOs who do not model this before launch get surprised at scale, with no clean way to recover margin.

CORE ELEMENTS

1. Infrastructure cost per customer (cloud + hosting, allocated by actual usage)
2. Support cost per customer (tickets x resolution time x cost per support hour)
3. Technology COGS component (for gross margin calculation and investor reporting)
4. AI feature cost per interaction, per user per month, and per product
5. Cost by customer segment (enterprise vs. SMB vs. free — often 10–20x different)
6. Gross margin by product line with full technology cost allocation applied
7. Cost trend over time (is cost per customer improving or degrading at scale?)

Tool 8: Unit Economics based cost allocator

Tool Snapshot

Technology Unit Economics Model

Quarterly review with CFO and Product. Updated as customer count, usage patterns, and infrastructure costs change. Critical input for pricing decisions.

Usage Notes

- Enterprise accounts cost 16x more to serve than SMB on tech — but generate 22x more revenue. The premium is justified.
- AI costs as % of gross margin are growing 15% QoQ. At this rate, AI is 8% of GM in 12 months — model this now and plan pricing accordingly.
- Free tier accounts cost \$2.10/month each to serve. With 12,000 free users, that is \$25K/month in unrecovered technology cost.
- Use this to justify tiered pricing: if enterprise AI and compliance features cost 5x more to deliver, pricing differential should reflect that.

Cost Component	Monthly Total	Per SMB Customer	Per Ent. Customer	Per Request	Notes
INFRASTRUCTURE					
Compute (cloud)	\$420K	\$4.20	\$84.00	\$0.00	Enterprise = 20x SMB usage avg.
Database & storage	\$180K	\$1.80	\$36.00	\$0.00	
CDN & networking	\$60K	\$0.60	\$12.00	\$0.00	
INFRA SUBTOTAL	\$660K	\$6.60	\$132.00	\$0.00	
AI / ML COSTS					
Foundation model API calls	\$112K	\$0.89	\$28.00	\$0.00	AI features active — 26% of customers
Vector DB & embeddings	\$28K	\$0.22	\$7.00	\$0.00	
AI SUBTOTAL	\$140K	\$1.11	\$35.00	\$0.01	
SUPPORT & OPERATIONS					
Customer support (eng. time)	\$85K	\$0.85	\$42.50	—	Enterprise gets 5x more eng. support hrs
Monitoring & incident response	\$40K	\$0.40	\$8.00	—	
Compliance & security overhead	\$35K	\$0.35	\$14.00	—	
SUPPORT SUBTOTAL	\$160K	\$1.60	\$64.50	—	
TOTAL TECH COST / CUSTOMER / MO	\$960K total	\$9.31/mo	\$231.50/mo	\$0.01	
REVENUE / CUSTOMER / MO		\$29.00	\$650.00		Blended ARPU
GROSS MARGIN (tech COGS only)		67.90%	64.40%		Target: 65%+ blended
AI COST AS % OF GROSS MARGIN		3.80%	5.40%		Growing 15% QoQ — model ahead



Tool 8: Unit Economics based cost allocator

Mini Case



Product Wants to Give AI Features to the Free Tier

Product proposes including AI-generated summaries in the free tier to drive activation. The feature costs \$0.89/user/month in API costs. The company has 12,000 free users. The CTO must evaluate financial viability before it reaches the roadmap.

09

Engineering Workforce Model



Tool 9: Engineering Workforce Cost Model

PERSPECTIVE

The 'just hire more engineers' reflex is increasingly expensive and slow — and increasingly unnecessary. AI coding tools, specialized offshore teams, managed services, and platform engineering have created a genuine spectrum of ways to add capacity, each with different cost profiles, ramp times, and quality tradeoffs.

CTOs who model workforce costs rigorously — comparing the true cost of FTEs against contractors, outsourcing, and AI-augmented capacity — make significantly better decisions than those who default to headcount.

In 2025, the AI productivity multiplier is the most significant new variable: the best engineering teams using AI coding tools are 30–50% more productive. Sometimes the right answer is not 'hire 5 engineers' but 'add AI tooling to your existing 10 engineers and recapture the equivalent of 4–5 FTEs without the headcount risk.'

CORE ELEMENTS

1. True FTE cost (salary $\times$ 1.4 benefits/overhead + \$30–50K recruiting + \$20K onboarding + management overhead)
2. Contractor / staff augmentation cost (hourly rate $\times$ hours — no benefits, but premium rate)
3. Outsourced team cost (by geography, specialization, management overhead required)
4. AI-augmented productivity value (capacity unlocked by AI tools vs. cost of tools)
5. Time-to-productivity for each model (ramp time before full contribution)
6. Quality and risk factors (IP ownership, security clearance, knowledge retention on exit)
7. Headcount planning model (build / buy / borrow / augment decision per skill)

Tool 9: Engineering Workforce Cost Model

Tool Snapshot

Engineering Workforce Cost Comparison Model

- Use to evaluate each capacity addition decision.
- The right model depends on the type of work — there is no single universal answer.

Usage Notes

- True FTE cost = salary x 1.4 + \$30–50K recruiting + \$20K onboarding. Commonly underestimated by 40%.
- AI coding tools at \$1,000/dev/month are the highest-ROI workforce line: 30% uplift on a \$200K engineer = \$60K value for \$12K cost.
- Never outsource core architecture decisions, security design, or customer data pipelines — these define your competitive moat.
- The offshore model works best when work is well-scoped and documented; it breaks when used as a shortcut for ambiguous product work.

Model	Monthly Cost (FTE equiv.)	Productivity	Best For	Key Risks	Annual Cost Range
FULL-TIME EMPLOYEE (FTE)					
Senior Engineer (US / Western Europe)	\$18,000–\$22,000	3–6 months	Core product, IP-sensitive, long-term work	Hiring takes 3–6 months; true cost 1.4x base salary	\$216K–\$264K
Senior Engineer (India / LATAM / E. Europe)	\$8,000–\$12,000	3–6 months	Core product, high-value feature work	Timezone management; requires documentation culture	\$96K–\$144K
CONTRACTORS / STAFF AUGMENTATION					
US-based senior contractor	\$22,000–\$28,000	4–8 weeks	Specific projects, surge capacity, niche skills	IP risk; no long-term commitment; costly at scale	\$264K–\$336K
Offshore senior contractor	\$6,000–\$10,000	4–8 weeks	Well-scoped work, QA, documented features	Quality variance; coordination overhead	\$72K–\$120K
MANAGED SERVICES / OUTSOURCED TEAMS					
Nearshore managed team (5-person)	\$30,000–\$45,000/team/mo	6–12 weeks	Non-core platforms, QA, support engineering	Quality governance needed; security review essential	\$360K–\$540K/team/yr
Infrastructure-as-a-Service / Platform	\$5,000–\$20,000/mo	Immediate	DevOps, infrastructure, monitoring	Vendor lock-in risk; SLA terms critical	\$60K–\$240K/yr
AI AUGMENTATION					
AI coding tools per developer (Cursor, Copilot)	\$500–\$1,000/dev/mo	1–2 weeks	All engineering — 30–50% productivity gain	Code review needed; data security policy required	\$6K–\$12K/dev/yr
10 engineers + AI tools (capacity output)	~13 FTE equivalent output	2 weeks to full uplift	Highest leverage for experienced senior teams	Requires senior engineers to supervise output	Tool cost: \$60K–\$120K/yr
STRATEGIC GUIDELINES					
Core platform, IP, AI supervision	FTE (premium market)				
Scalable feature delivery, growth work	FTE (distributed) or staff aug.				
QA, testing, documentation	Managed service or offshore				
ALL engineering work	Layer AI tools as a multiplier — non-negotiable				



Tool 9: Engineering Workforce Cost Model Mini Case



Should We Hire 10 Engineers or Find Another Way?

The product roadmap requires 10 additional engineering FTEs by Q3. At blended \$220K/FTE/year, that is \$2.2M annually plus \$400K in recruiting and 4–6 months to ramp. The CFO questions whether there is a faster, cheaper path to the same output.

10

Technology Risk Register (Financially Quantified)



Tool 10: Technology Risk Register (Financially Quantified)

PERSPECTIVE

Cybersecurity budgets, resilience investments, and disaster recovery programs all compete for funding against features and growth. Without a financial framing, they almost always lose — 'we might get hacked' is not a budget argument. 'There is a 25% annual probability of a data breach costing \$8.5M, compared to a \$420K mitigation investment' is.

A financially quantified risk register changes the conversation: Expected Annual Loss (probability × impact) gives the CFO a model they can work with, and the ROI of risk mitigation becomes directly comparable to the ROI of revenue-generating projects.

In 2025, three new risk dimensions are live: AI-specific risks (model poisoning, hallucination in production, data leakage via prompts), third-party supply chain risks (AI API providers becoming critical dependencies), and regulatory risk (EU AI Act, US state privacy laws now affecting technology architecture).

CORE ELEMENTS

1. Risk inventory (technical, operational, security, compliance, AI-specific)
2. Annual probability of occurrence (assessed from incident history and industry data)
3. Impact quantification in dollars (remediation + fines + legal fees + churn cost)
4. Expected Annual Loss — EAL (probability × impact = financially comparable metric)
5. Mitigation cost (investment that reduces probability or impact, and by how much)
6. Residual risk (EAL remaining after the mitigation investment is made)
7. Risk appetite statement (the level of EAL the board has explicitly approved accepting)
8. Regulatory risk mapping (GDPR, SOC2, EU AI Act, SEC cyber disclosure rules)

Tool 10: Technology Risk Register (Financially Quantified)

Tool Snapshot

Technology Risk Register with Financial Quantification

Reviewed quarterly with CISO and CFO. Top 5 risks presented to board annually. Used to justify security and resilience budget requests.

Usage Notes

- Impact figures include: remediation costs, regulatory fines, customer compensation, legal fees, and churn impact (revenue × churn rate × LTV).
- The 3.5:1 ROI on the mitigation program is the board-level argument for the security and resilience budget.
- AI-specific risks are the fastest-growing risk category in 2025 — review monthly, not quarterly.
- EAL is a decision-making framework, not a spending forecast. Use it to prioritize mitigation investments, not to budget for incident costs.

Risk	Category	Annual Probability	Impact if Occurs	Expected Annual Loss	Mitigation Cost	EAL After Mitigation	Priority
Data breach — customer PII (external attack)	Security	25%/yr	\$8.5M	\$2,125K	\$420K	\$280K	P0
Critical system outage exceeding 4 hours	Resilience	35%/yr	\$2.4M	\$840K	\$180K	\$120K	P0
Ransomware / extended business interruption	Security	15%/yr	\$6.2M	\$930K	\$280K	\$185K	P1
SOC2 Type II audit failure (compliance gap)	Compliance	20%/yr	\$1.8M	\$360K	\$140K	\$60K	P1
AI model producing harmful output in production	AI / ML Risk	40%/yr	\$1.2M	\$480K	\$95K	\$120K	P1
Critical vendor failure (key API dependency)	3rd-Party Risk	10%/yr	\$3.5M	\$350K	\$60K	\$35K	P2
EU AI Act non-compliance (automated decisions)	Regulatory	30%/yr	\$2.8M	\$840K	\$220K	\$140K	P1
Financial data pipeline failure (reporting)	Operational	25%/yr	\$1.1M	\$275K	\$80K	\$40K	P2
TOTAL PORTFOLIO				\$6,200K EAL	\$1,475K invest.	\$980K residual	
NET VALUE OF MITIGATION PROGRAM				Reduces EAL by \$5.22M	for \$1.475M	3.5:1 expected ROI	



Tool 10: Technology Risk Register (Financially Quantified)

PERSPECTIVE

Cybersecurity budgets, resilience investments, and disaster recovery programs all compete for funding against features and growth. Without a financial framing, they almost always lose — 'we might get hacked' is not a budget argument. 'There is a 25% annual probability of a data breach costing \$8.5M, compared to a \$420K mitigation investment' is.

A financially quantified risk register changes the conversation: Expected Annual Loss (probability × impact) gives the CFO a model they can work with, and the ROI of risk mitigation becomes directly comparable to the ROI of revenue-generating projects.

In 2025, three new risk dimensions are live: AI-specific risks (model poisoning, hallucination in production, data leakage via prompts), third-party supply chain risks (AI API providers becoming critical dependencies), and regulatory risk (EU AI Act, US state privacy laws now affecting technology architecture).

CORE ELEMENTS

1. Risk inventory (technical, operational, security, compliance, AI-specific)
2. Annual probability of occurrence (assessed from incident history and industry data)
3. Impact quantification in dollars (remediation + fines + legal fees + churn cost)
4. Expected Annual Loss — EAL (probability × impact = financially comparable metric)
5. Mitigation cost (investment that reduces probability or impact, and by how much)
6. Residual risk (EAL remaining after the mitigation investment is made)
7. Risk appetite statement (the level of EAL the board has explicitly approved accepting)
8. Regulatory risk mapping (GDPR, SOC2, EU AI Act, SEC cyber disclosure rules)

Tool 10: Technology Risk Register (Financially Quantified)

Tool Snapshot

Technology Risk Register with Financial Quantification

Reviewed quarterly with CISO and CFO. Top 5 risks presented to board annually. Used to justify security and resilience budget requests.

Usage Notes

- Impact figures include: remediation costs, regulatory fines, customer compensation, legal fees, and churn impact (revenue × churn rate × LTV).
- The 3.5:1 ROI on the mitigation program is the board-level argument for the security and resilience budget.
- AI-specific risks are the fastest-growing risk category in 2025 — review monthly, not quarterly.
- EAL is a decision-making framework, not a spending forecast. Use it to prioritize mitigation investments, not to budget for incident costs.

Risk	Category	Annual Probability	Impact if Occurs	Expected Annual Loss	Mitigation Cost	EAL After Mitigation	Priority
Data breach — customer PII (external attack)	Security	25%/yr	\$8.5M	\$2,125K	\$420K	\$280K	P0
Critical system outage exceeding 4 hours	Resilience	35%/yr	\$2.4M	\$840K	\$180K	\$120K	P0
Ransomware / extended business interruption	Security	15%/yr	\$6.2M	\$930K	\$280K	\$185K	P1
SOC2 Type II audit failure (compliance gap)	Compliance	20%/yr	\$1.8M	\$360K	\$140K	\$60K	P1
AI model producing harmful output in production	AI / ML Risk	40%/yr	\$1.2M	\$480K	\$95K	\$120K	P1
Critical vendor failure (key API dependency)	3rd-Party Risk	10%/yr	\$3.5M	\$350K	\$60K	\$35K	P2
EU AI Act non-compliance (automated decisions)	Regulatory	30%/yr	\$2.8M	\$840K	\$220K	\$140K	P1
Financial data pipeline failure (reporting)	Operational	25%/yr	\$1.1M	\$275K	\$80K	\$40K	P2
TOTAL PORTFOLIO				\$6,200K EAL	\$1,475K invest.	\$980K residual	
NET VALUE OF MITIGATION PROGRAM				Reduces EAL by \$5.22M	for \$1.475M	3.5:1 expected ROI	



Tool 10: Technology Risk Register (Financially Quantified) Mini Case



Getting \$1.5M in Cybersecurity Budget Approved

The CISO and CTO jointly need \$1.5M in incremental security investment for FY2026. The CFO's previous response to security budget requests: 'We have not been breached, so clearly what we have is working.' The board has just asked for a risk report following a high-profile breach at a direct competitor.



Tool 10: Technology Risk Register (Financially Quantified)

Mini Case Solution Playbook

Build the register with company-specific numbers

Use your company's actual revenue, customer count, and contractual breach penalties — not generic industry statistics. The IBM Cost of Data Breach Report provides base figures; adjust for your specific regulatory exposure and customer data profile.

Lead with Expected Annual Loss, not probability

Replace the CFO's mental model ('we have not been breached, so risk must be low') with EAL: 'Our current Expected Annual Loss from technology risk is \$6.2M. We are proposing \$1.475M that reduces our EAL to \$980K — a \$5.2M reduction for \$1.5M spent.'

Use the competitor breach as a calibration anchor

The competitor's breach cost is likely in public SEC filings. 'If the same attack hit us, our breach would cost approximately \$X because of our customer data profile and EU regulatory exposure.' This grounds the model in real-world events.

Structure as three tranches with risk reduction milestones

Tranche 1 (\$420K): addresses P0 breach risk. Tranche 2 (\$600K): addresses P1 ransomware and compliance. Tranche 3 (\$480K): addresses AI and regulatory risks. Each tranche has a specific EAL metric it improves.

Outcome

CFO approves the full \$1.5M with one condition: an annual risk register showing EAL reduction against the investment. Board adds technology risk to the quarterly governance agenda. CTO now has a financial framework to sustain security investment year over year.

Thank You